Data Analysis and Storytelling

Regression analysis - model selection

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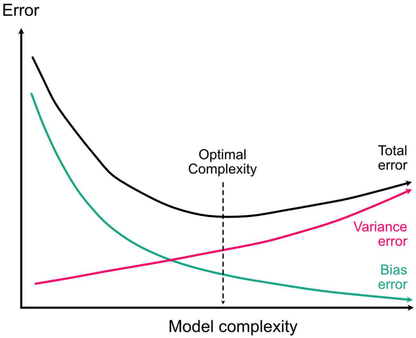
FIRS - Track 1

Selecting the right model

- We saw that just with a basic lm, we already have quite some room in model buidling: (i) multiple variables, (ii) interactions, (iii) variable transformation
- Even before seeing further extensions such as (i) other distributions, (ii) hierarchical models or (iii) space/time structure, we have already a potentially very large number of models that can be fitted
- How to select the right model?

Bias-variance trade-off

- The more covariables/complexity we add to a model, the better it can adjust to the true relation, reducing the bias
- The more covariables/complexity we add to a model, the larger the error on the estimated coefficients, increasing model variance
- Too few covariables/complexity and we are underfitting, too many and we are overfitting, this is the bias-variance trade-off



Aim of the model

The data analyst usually wants a model either to explain (understand effects) or to predict.
For instance:

- What is the effect of climate change on tree mortality? [explain or predict]
- What is the projected rate of tree mortality with future expected climate change? [explain or predict]
- Are strict forest reserve more resilient than managed forests? [explain or predict]
- How is forest biomass varying across France? [explain or predict]

Model selection and prediction

Because of the bias-variance trade-off fitting the most complex we can is not optimal since we increase prediction uncertainty.
Many options exists to do model selection to optimize model prediction capacities, the two most interesting are:

1. AIC selection
2. Shrinkage estimation

AIC model selection

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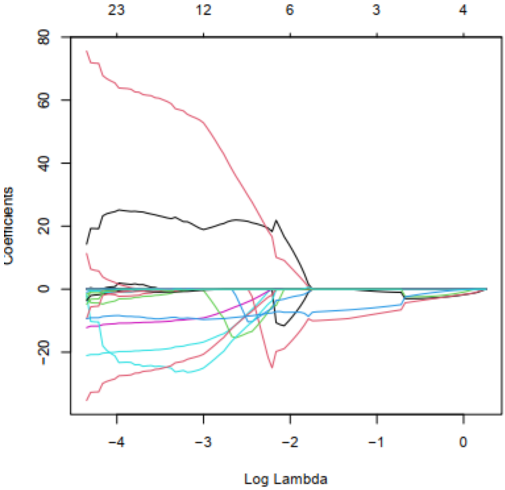
- Lower AIC means better models, we select the model we the lowest AIC value
- AIC is assymptotically similar to leave-one-out cross-validation, AIC optimize the predictive error of the model on new data

Shrinkage estimator

- The idea behind shrinkage estimator is that among many potential covariables, a few only are important to predict the response.
- Shrinkage puts a penalty (λ) on regression coefficients, pulling them towards 0, there are two main options: (i) LASSO or (ii) ridge regression



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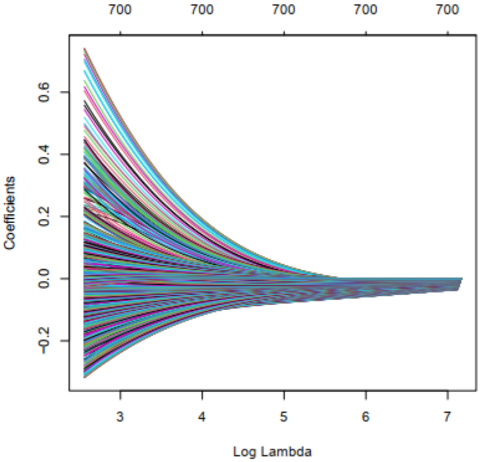

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- LASSO does automatic model selection, penalized variables are shrunk to 0 - sparse estimator
- Ridge shrinks regression coefficients towards 0 - not sparse estimator but better for $k >> n$ problems
- The penalty term λ must be optimized, for instance via cross-validation

p-hacking

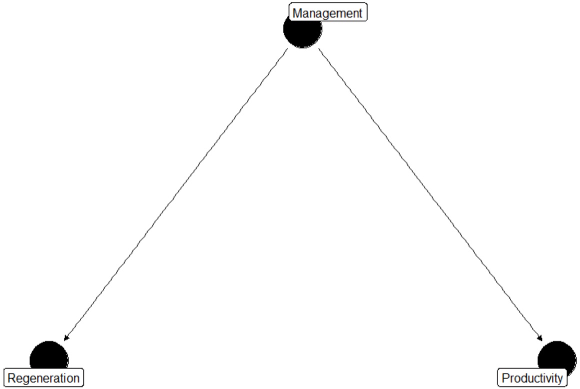
Model selection should **not** be done by looking at p-values, this is called p-hacking and is considered scientific misconduct since by tweaking certain model components or dropping observations, one can almost always get the desired significant p-values.
Also, with large enough sample size, any statistical tests will turn up significant even for *insignificant* effect sizes.

Variable selection and explanation - a warning

- We saw in the previous lecture that with collinearity between variables, building a model dropping some variables leads to biased estimation of effect sizes.
- As a result data-driven model selection approaches will wrongly estimate effect sizes and confidences intervals.
- So how to build model when the aim is to understand/explain - aka to do inference?

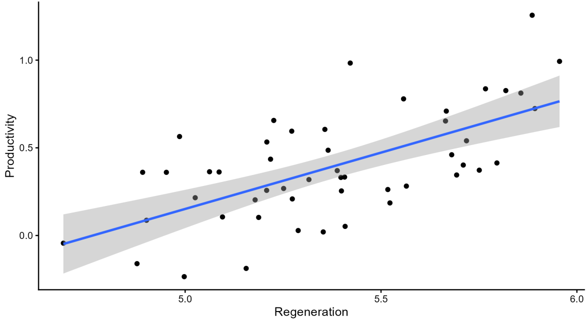
Graphs to the rescue

Say that we are interested in estimating the effect of management on adult tree productivity, yet management also affect (for whatever reason) seedling abundance. We have the following graph:



Spurious correlation

In this virtual exemple, regeneration and productivity are not related to each other, yet because they share a common cause, they are correlated:



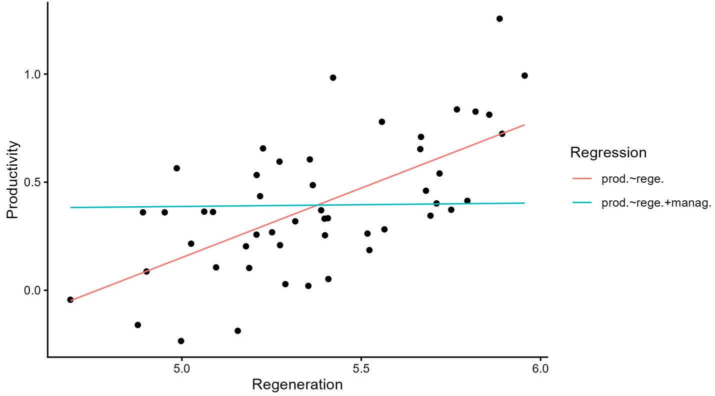
If a model *productivity* $\sim$ *regeneration* is fitted, one could wrongly conclude for an effect of regeneration on productivity.

Experimental control

- If we did an experiment with constant management and varying regeneration level, we would get the correct flat relation between regeneration and productivity
- Experimental design aims at controlling all potential confounders to focus on the effect under study
- But what if we cannot run an experiment?

Regression models to the help of causal inference

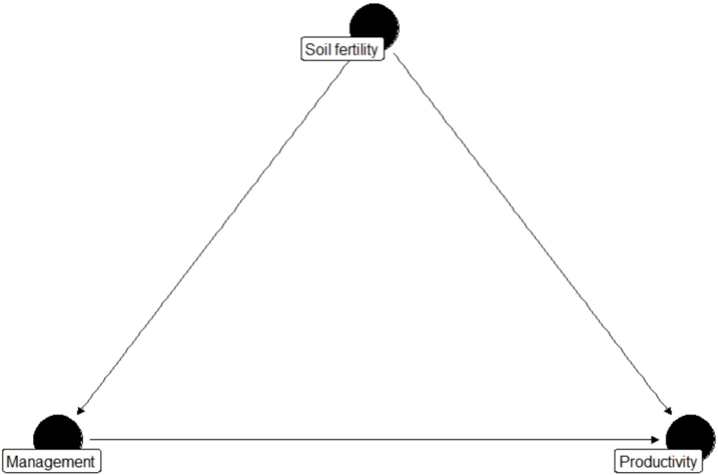
- Regression models are a tool to do inference on observational data
- Remember that the regression coefficients in multivariate models are the estimated individual covariable effects while holding all other covariables constant.



Confounders, colliders and mediators

There are three basic type of structure in graphs: (i) confounders, (ii) colliders and (iii) mediators.

- Confounder: is a covariable affecting both the variable of interest (management) and the response (productivity). Confounders should always be included in the model.



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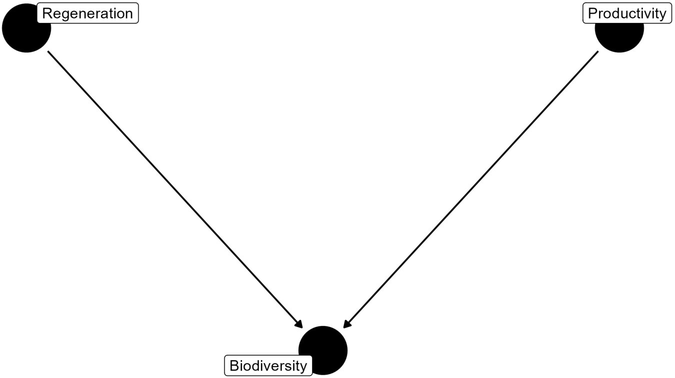
- Collider: a covariable linked to both the variable of interest (management) and by the response (productivity). A collider should **not** be included in the model.



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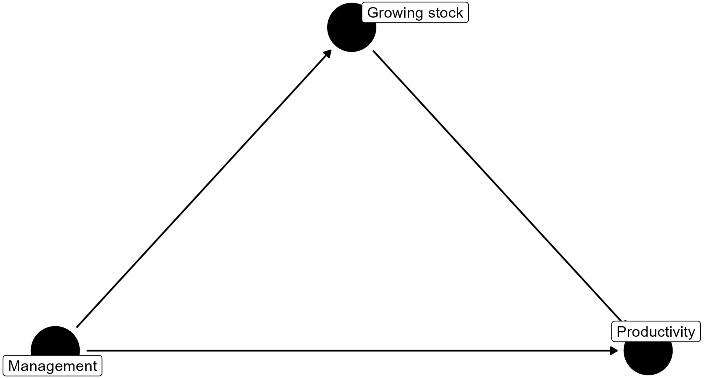
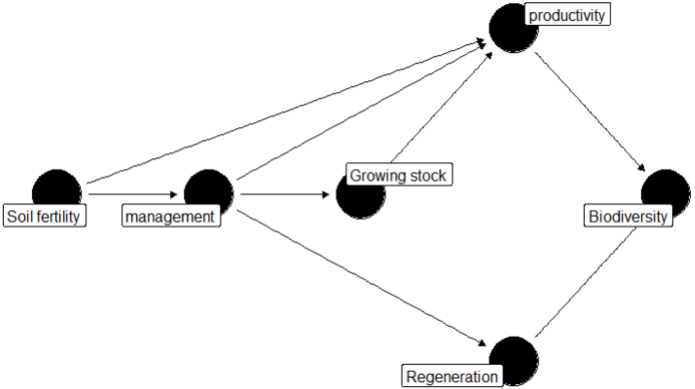
- Mediator: a covariable affected by the variable of interest (management) and affecting the response (productivity). Depending on the aim of the study a mediator can be included or not.



Graph to the rescue

- To select a model to understand a given relation we therefore rely on building graphs to identify variables to include in the model
- Given an identified relation (management $->$ productivity), we add all confounders, we do not add colliders and depending on the aim we add mediators.



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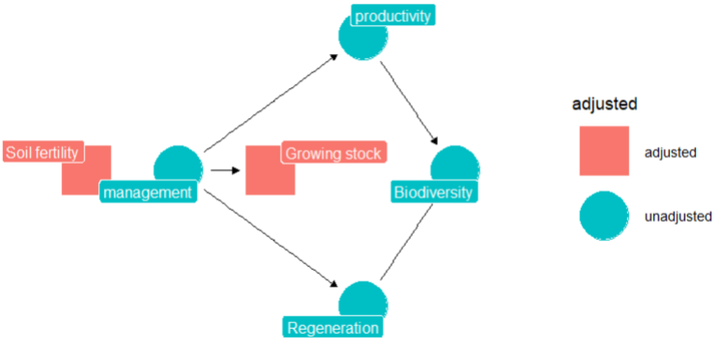

$$Productivity \sim Management + Fertility + Growing\ stock$$

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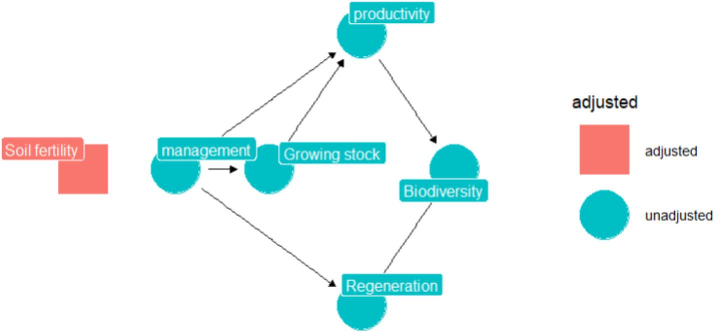

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- If the aim of the analysis is prediction, (semi-)automatic methods such as AIC or shrinkage estimation are appropriate
- If the aim of the analysis is explanation, we can either rely on experimental design or use regression with observationnal data
- With observationnal data, building conceptual graphs around the targeted relation helps identify the variable to include in the model